Requirements Coverage Report

COMP3211 Group 149 PIM | Actual handoff status | 20 October 2026

US1-US11/UR1-UR11 and FR1-FR36 are implemented and verified locally under the qualified stored-alarm scope. Windows acceptance, the real MP4 and the official completed declaration remain outstanding; this is not a complete submission.

Sources: Project Description - 2026, pp. 1-4; selected SRS v1.3; merged Design v1.1; implementation plan. Handoff SRS v1.4/Design v1.2 synchronize only the approved EOF clarification and SRS reference.

Actual local run: CPython 3.12.6, macOS 15.7.3 arm64, 249 passing methods. Logs: docs/evidence/LOCAL_VERIFICATION.txt. Implemented/local pass means executed local evidence; partial/unverified needs platform evidence; Q1 needs instructor confirmation; not provided means absent. E IDs index actual code/tests below.

User stories and SRS user requirements

Appendix B user stories map one-to-one to SRS section 3.1 user requirements.

US / UR	Required behaviour	Status	Code/test evidence
US1 / UR1	UR1. The user shall be able to manage plain-text, task, event and contact PIRs in one collection.	Implemented / local	E01
US2 / UR2	UR2. The user shall be able to create a plain-text PIR containing a note.	Implemented / local	E02, E06
US3 / UR3	UR3. The user shall be able to create a task PIR containing a description and deadline.	Implemented / local	E03
US4 / UR4	UR4. The user shall be able to create an event PIR containing a description, starting time and alarm time.	Implemented / local; Q1 qualified	E04
US5 / UR5	UR5. The user shall be able to create a contact PIR containing a name, address and mobile number.	Implemented / local	E05, E06
US6 / UR6	UR6. The user shall be able to modify the business fields of an existing PIR.	Implemented / local	E07, E11
US7 / UR7	UR7. The user shall be able to search by type, text and time, including criteria combined with &&, || and !.	Implemented / local	E08, E09, E10, E11, E12, E13
US8 / UR8	UR8. The user shall be able to display detailed information about one PIR or all PIRs.	Implemented / local	E14
US9 / UR9	UR9. The user shall be able to delete a specified PIR.	Implemented / local	E15
US10 / UR10	UR10. The user shall be able to save the current PIR collection to a .pim file.	Implemented / local	E16, E18

US / UR	Required behaviour	Status	Code/test evidence
US11 / UR11	UR11. The user shall be able to load a saved PIR collection and continue working with it.	Implemented / local	E17, E18

Functional requirements

All 36 FRs below are implemented and passed local verification. Public interfaces and MVC allocation were independently audited; qualified Windows acceptance is reported separately, not silently marked passed.

SRS ID	Specified behaviour	Status	Evidence
FR1	One initially empty collection, no automatic load	Yes; local	E01, E21
FR2	New ID above session maximum allocated/loaded; no counter decrease/failed allocation	Yes; local	E01, E21
FR3	Exactly four supported types; unique collection IDs	Yes; local	E01, E22, E25
FR4	Valid PLAIN_TEXT creation	Yes; local	E02, E23
FR5	Valid TASK creation	Yes; local	E03, E23, E25
FR6	Valid EVENT creation, alarm no later than start	Yes; local	E04
FR7	Valid CONTACT; mobile stored as text	Yes; local	E05
FR8	Reject complete invalid creation without state/ID change	Yes; local	E02, E03, E04, E05
FR9	Update one business field with whole-record validation; fixed ID/type	Yes; local	E07, E11
FR10	Invalid/absent ID or invalid edit preserves all records	Yes; local	E07
FR11	Direct/saved search of all current records, without mutation	Yes; local	E08, E11
FR12	type = X exact match	Yes; local	E08
FR13	Case-sensitive literal non-empty text substring, untrimmed	Yes; local	E09
FR14	Full minute-precision <, >, = on time fields	Yes; local	E09
FR15	!, &&, ||, parentheses and precedence	Yes; local	E10, E11
FR16	Missing field atomic false; negation uses that result	Yes; local	E10
FR17	Invalid expression/criterion yields error and no result set/mutation	Yes; local	E10, E11
FR18	Search shows complete records in numeric ID order; zero matches explicit	Yes; local	E12
FR19	Display one complete record or ID error	Yes; local	E14
FR20	Display all ordered records or explicit empty collection	Yes; local	E14
FR21	Display selected record before deletion confirmation; refusal preserves it	Yes; local	E15

SRS ID	Specified behaviour	Status	Evidence
FR22	Save entire sorted collection, including empty, to lowercase .pim; exclude criteria	Yes; local	E16
FR23	Failed save preserves collection/counter/dirty; success only after write	Yes; local	E16
FR24	Fully validated load replaces exactly, restores IDs; empty replacement valid	Yes; local	E17
FR25	Reject missing/unreadable/invalid file before commit; retain state except prior successful save	Yes; local	E17
FR26	Preserve number/IDs/types/values incl. Unicode/mobile zeroes; accept duplicate contents	Yes; local	E18
FR27	Text commands/arguments for operations, no numbered-menu dependency	Yes; local	E19
FR28	Unknown commands/bad arguments error then usable prompt; no :cancel mechanism	Yes; local	E19
FR29	Explicit exit respecting unsaved guards	Yes; local	E20
FR30	Define/list immutable sequential session criteria; invalid definition adds nothing	Yes; local	E13
FR31	Criteria survive changes/loads, evaluate current collection; not persisted	Yes; local	E11, E13
FR32	Normalized same-type duplicate create warning/confirmation, excluding ID	Yes; local	E06
FR33	Confirm overwriting existing file; refusal preserves file and is not successful save	Yes; local	E16
FR34	Dirty only on successful create/delete/changed edit; clear after startup/save/load	Yes; local	E20
FR35	Dirty load/exit offers save/discard/cancel; do not discard old data before valid load	Yes; local	E20
FR36	Save-and-continue proceeds only after success; failed save/refused overwrite stops action	Yes; local	E20

Nonfunctional and common requirements

SRS ID	Specified behaviour	Status	Evidence / limit
NFR1	NFR1. Every error shall begin with ERROR and include a short reason identifying the affected input or operation. Where relevant, feedback shall state the expected format or allowed values.	Implemented; local pass	E21
NFR2	NFR2. Invalid commands, fields, criteria and expected file-operation errors shall not terminate the program through an unhandled exception. Validation and load failures shall preserve state as specified in FR8, FR10, FR17 and FR25.	Implemented; local pass	E21
NFR3	NFR3. All code belonging to the model shall reside in model and shall be callable independently of terminal input and output. This includes any persistence code classified as model behaviour. Command parsing, prompting and output formatting shall remain separately identifiable.	Implemented; local pass	E22, E25
NFR4	NFR4. The system shall run on Windows with the Python version identified in the developer manual, without third-party runtime Python packages.	Partial; Windows unverified	E23 Actual Windows execution remains pending.
NFR5	NFR5. All specified record, search and file operations shall work without a network connection.	Implemented; local pass	E23, E25 Actual Windows execution remains pending.
S 3.2/3.3	Offline terminal interaction, standard-library runtime, all model code in model, automated model tests.	Implemented; local pass	15 runtime modules audited; model/CLI catalog below; formal Testing report separate.
S 5.3	Exact fields, single-line Unicode, normalization, mobile/date rules, literal escaping, stdin/stdout, lowercase .pim.	Implemented; local pass	E01-E21; model/CLI/persistence tests; Q6 exception below.
S 1.4 / Q1	Alarm is a stored/manageable/searchable time; alarm <= start; no automatic notifications.	Implemented as qualified baseline	E04 and Event tests; instructor confirmation outstanding.
Q6 clarification	Dirty permanent EOF: stderr warning/status 1; no repeated prompt, autosave, overwrite or implicit discard; input-only Ctrl-C recovery.	Implemented; local/process pass	Final catalog EOF row; synchronized in handoff SRS 5.3 and Design 8.

Verification and limits

The exact Design Figure 5 scenario was executed through model APIs and real main.py. Tasks 2 and 7 initially matched type/text/deadline; after updating only Task 2's deadline to 2026-11-25 23:59, direct and saved searches returned only 7. Task 7 remained unchanged. Reloading restored current results using the same saved criterion. Additional regressions verify long symbolic expressions, invalid skipped branches and ID history after deletion/lower loads.

No additional required record operation is known unimplemented under the approved baseline. Automatic alarm notifications are intentionally excluded by the provisional SRS assumption, not claimed instructor-approved.

Windows, actual permissions/terminal behaviour there, and its offline acceptance procedure have not been executed. Private long-expression evaluation was fixed without changing the public grammar or interfaces. Extremely deep parser nesting returns a controlled ExpressionError. Concurrent writers and power-loss recovery remain outside the specified guarantee.

Assignment Implementation and related deliverables

Item	Actual artifact	Status	Remaining action
Implementation 3.a	main.py, model/, cli/, tests/	Prepared; locally verified	Real Windows verification and student review
Implementation 3.b	docs/DEVELOPER_MANUAL.pdf	Prepared in allowed PDF format	Execute/record the unverified Windows and optional IDE steps
Implementation 3.c	docs/USER_MANUAL.pdf	Prepared in allowed PDF format	Student review; Windows platform confirmation
Implementation 3.d	docs/DEMO_RECORDING_GUIDE.pdf and demo_commands.txt	Actual MP4 NOT PROVIDED	Students record a real MP4 <=4 minutes; script does not replace it
Implementation 3.e	Root REQUIREMENTS_COVERAGE.pdf (this report)	Prepared in allowed PDF format	Update only if actual code/platform evidence changes
Separate Testing 4	Model tests; root MODEL_LINE_COVERAGE.pdf	Prepared; measured locally	Keep requirements coverage distinct from measured line coverage
Official declaration	docs/AI_ASSISTANCE_SUMMARY.pdf supports official form	Official completed form NOT PROVIDED	Students supply genuine identities/contributions/signatures at ZIP root

Code and actual verification catalog

Each E row below gives production allocation and real test method references. All referenced methods were executed in the passing suite. The existing plan retains finer state-transition and source-section details. Code is relative to the source root; abbreviated component names resolve through the Developer Manual architecture table. These references are traceability evidence, not a claim that every possible input or every branch has been tested.

ID	Production modules	Actual executed test references
E01	model/records.py, model/service.py, main.py	tests/test_records.py::test_exact_record_types_and_constructor_fields; tests/test_service.py::test_empty_startup_without_store_access; tests/test_service.py::test_deleting_highest_or_all_records_does_not_reuse_ids; tests/test_persistence.py::test_load_higher_lower_and_empty_ids_never_lowers_session_high_water; tests/test_entrypoint.py::test_startup_never_loads_existing_sample_or_previous_files
E02	validation, records, service	tests/test_validation.py::test_outer_whitespace_removed_internal_unicode_preserved; tests/test_validation.py::test_empty_blank_and_underscore_only_rejected; tests/test_service.py::test_invalid_creation_does_not_consume_ids_or_change_dirty

ID	Production modules	Actual executed test references
E03	validation, records, service	tests/test_validation.py::test_normal_past_leap_and_extreme_years; tests/test_validation.py::test_invalid_calendar_and_clock_boundaries; tests/test_validation.py::test_missing_extra_id_and_type_fields_rejected
E04	validation, records, service	tests/test_records.py::test_event_alarm_before_equal_and_after_start; tests/test_service.py::test_complete_event_candidate_revalidates_alarm_and_start_atomically
E05	validation, records, service	tests/test_validation.py::test_plus_zeroes_and_digit_boundaries_preserved; tests/test_validation.py::test_invalid_syntax_and_non_ascii_digits_rejected; tests/test_records.py::test_direct_contact_validates_mobile_syntax
E06	records, service, controller/view	tests/test_service.py::test_duplicates_use_all_normalized_fields_and_report_sorted_ids; tests/test_service.py::test_updates_can_produce_duplicate_content_without_prompting; tests/test_controller.py::test_duplicate_refusal_does_not_consume_id_or_change_dirty; tests/test_controller.py::test_duplicate_invalid_response_then_confirmation_allocates_new_id
E07	service, records, validation, CLI parser	tests/test_service.py::test_update_each_business_field_preserves_other_data_id_and_type; tests/test_service.py::test_invalid_field_values_leave_original_and_dirty_unchanged; tests/test_service.py::test_noop_updates_preserve_original_identity_and_both_dirty_states; tests/test_cli_parser.py::test_update_requires_exactly_one_assignment
E08	criteria, search, service	tests/test_criteria.py::test_type_matches_each_supported_type; tests/test_service.py::test_search_uses_all_current_records_without_mutation
E09	criteria, search, validation	tests/test_criteria.py::test_substring_is_preserved_without_trimming_or_unicode_normalization; tests/test_criteria.py::test_all_time_fields_and_comparisons; tests/test_search.py::test_escaped_quotes_backslashes_and_no_double_decoding
E10	criteria, search, service	tests/test_criteria.py::test_boolean_truth_tables; tests/test_search.py::test_comparison_not_and_or_precedence_in_ast; tests/test_search.py::test_validate_all_branches_before_any_record_evaluation; tests/test_search.py::test_missing_field_negation_applies_after_atomic_false; tests/test_service.py::test_invalid_search_is_validated_even_when_collection_is_empty
E11	criteria, service, controller/view	tests/test_model_regressions.py::test_long_flat_conjunction_works_directly_and_as_saved_criterion; tests/test_model_regressions.py::test_long_disjunction_and_negation_keep_boolean_semantics; tests/test_model_regressions.py::test_invalid_branch_after_long_valid_prefix_is_still_fully_validated; tests/test_model_regressions.py::test_design_search_then_update_scenario_and_reload; tests/test_entrypoint.py::test_exact_design_figure5_search_update_and_next_search; tests/test_entrypoint.py::test_long_valid_expression_and_invalid_branch_keep_cli_usable
E12	service, controller, view	tests/test_view.py::test_complete_records_in_numeric_order_and_canonical_time_format; tests/test_controller.py::test_clean_startup_empty_messages_and_normal_exit; tests/test_controller.py::test_show_and_search_display_numeric_ids_and_every_loaded_field

ID	Production modules	Actual executed test references
E13	search, service, persistence, controller/view	tests/test_search.py::test_invalid_definitions_consume_no_id_or_modify_existing_entries; tests/test_search.py::test_numeric_id_allocation_and_listing_beyond_c9; tests/test_service.py::test_saved_criteria_reevaluate_after_create_update_and_delete; tests/test_persistence.py::test_saved_criteria_survive_load_and_reevaluate_loaded_values; tests/test_controller.py::test_complete_create_update_show_and_direct_saved_search
E14	service, controller, view	tests/test_service.py::test_absent_ids_report_not_found_without_mutation; tests/test_view.py::test_complete_records_in_numeric_order_and_canonical_time_format; tests/test_controller.py::test_clean_startup_empty_messages_and_normal_exit
E15	service, controller/view	tests/test_controller.py::test_delete_displays_before_confirmation_then_selected_only_mutation; tests/test_controller.py::test_delete_refusal_and_missing_id_do_not_call_delete
E16	persistence, service, controller/view	tests/test_persistence.py::test_exact_schema_encoding_and_fixed_key_order; tests/test_persistence.py::test_empty_collection_is_a_valid_round_trip; tests/test_persistence.py::test_every_expected_save_failure_phase_preserves_service_state; tests/test_persistence.py::test_replace_failure_preserves_existing_or_absent_target_and_cleans_temp; tests/test_controller.py::test_overwrite_refusal_preserves_target_dirty_and_reports_no_success; tests/test_controller.py::test_overwrite_invalid_response_then_confirmed_retry
E17	persistence, service, controller	tests/test_persistence.py::test_invalid_and_noncanonical_business_values_are_not_repaired; tests/test_persistence.py::test_duplicate_json_keys_at_root_and_record_levels_are_rejected; tests/test_persistence.py::test_duplicate_record_ids_are_rejected; tests/test_persistence.py::test_invalid_last_record_preserves_all_state_in_clean_and_dirty_cases; tests/test_persistence.py::test_load_replaces_ids_without_merging_or_reallocation; tests/test_controller.py::test_failed_load_after_discard_retains_original_dirty_records; tests/test_controller.py::test_failed_load_after_successful_preliminary_save_retains_clean_records
E18	persistence, service	tests/test_persistence.py::test_all_types_round_trip_preserves_exact_values_and_numeric_ids; tests/test_persistence.py::test_duplicate_business_content_under_distinct_ids_is_valid; tests/test_persistence.py::test_duplicate_contents_load_without_creation_confirmation
E19	CLI literals/parser/controller/view	tests/test_cli_parser.py::test_all_command_forms_and_argument_types; tests/test_cli_parser.py::test_bad_assignment_separators_quotes_and_escapes; tests/test_cli_parser.py::test_expression_tail_preserves_quotes_backslashes_and_trailing_spaces; tests/test_controller.py::test_expected_errors_recover_and_include_cause_and_guidance; tests/test_view.py::test_confirm_invalid_responses_retry_without_implied_consent
E20	service, controller/view	tests/test_controller.py::test_dirty_exit_cancel_returns_to_command_prompt; tests/test_controller.py::test_dirty_load_cancel_stops_before_file_access; tests/test_controller.py::test_exit_save_success_and_discard_are_distinct; tests/test_controller.py::test_failed_preliminary_save_stops_load_or_exit_and_allows_later_discard; tests/test_controller.py::test_declined_preliminary_overwrite_stops_load_or_exit; tests/test_controller.py::test_save_then_load_successful_transaction_sequence

ID	Production modules	Actual executed test references
E21	errors, parser/controller/view, persistence	tests/test_controller.py::test_expected_errors_recover_and_include_cause_and_guidance; tests/test_view.py::test_error_prefix_exactly_once_and_normal_output_uses_stdout; tests/test_entrypoint.py::test_invalid_input_recovers_without_traceback
E22	Every model module, main.py, CLI	tests/test_service.py::test_service_operations_never_use_terminal_or_storage; tests/test_persistence.py::test_real_save_load_do_not_use_terminal_io
E23	All runtime modules	Actual source/import review; platform limitations stated above.
E24	controller, private view stderr helper, main.py	tests/test_controller.py::test_dirty_permanent_eof_at_all_prompts_never_mutates_or_loops; tests/test_controller.py::test_recoverable_interrupt_at_each_input_boundary_preserves_state; tests/test_controller.py::test_keyboard_interrupt_from_model_commit_is_not_caught_and_retried; tests/test_entrypoint.py::test_dirty_eof_at_command_and_all_confirmation_boundaries_is_nonzero; tests/test_entrypoint.py::test_clean_eof_status_and_eof_after_successful_save
E25	model/service.py, cli/, main.py	tests/test_model_regressions.py::test_full_model_workflow_without_terminal_or_network; tests/test_entrypoint.py::test_complete_cli_workflow_with_socket_creation_blocked

Requirements coverage versus code coverage

The separate measured model result is 659/661 executable lines (99.70%), using 175 model tests only. The two uncovered lines and bytecode/trace method are in MODEL_LINE_COVERAGE.pdf. No minimum percentage is prescribed. Neither that percentage nor the number of passing tests substitutes for the US/UR/FR/NFR implementation status reported here.

The submission is not complete while the real implementation MP4, Windows acceptance and genuine official declaration remain outstanding. Separate presentation artifacts and final ZIP completeness are still student tasks.